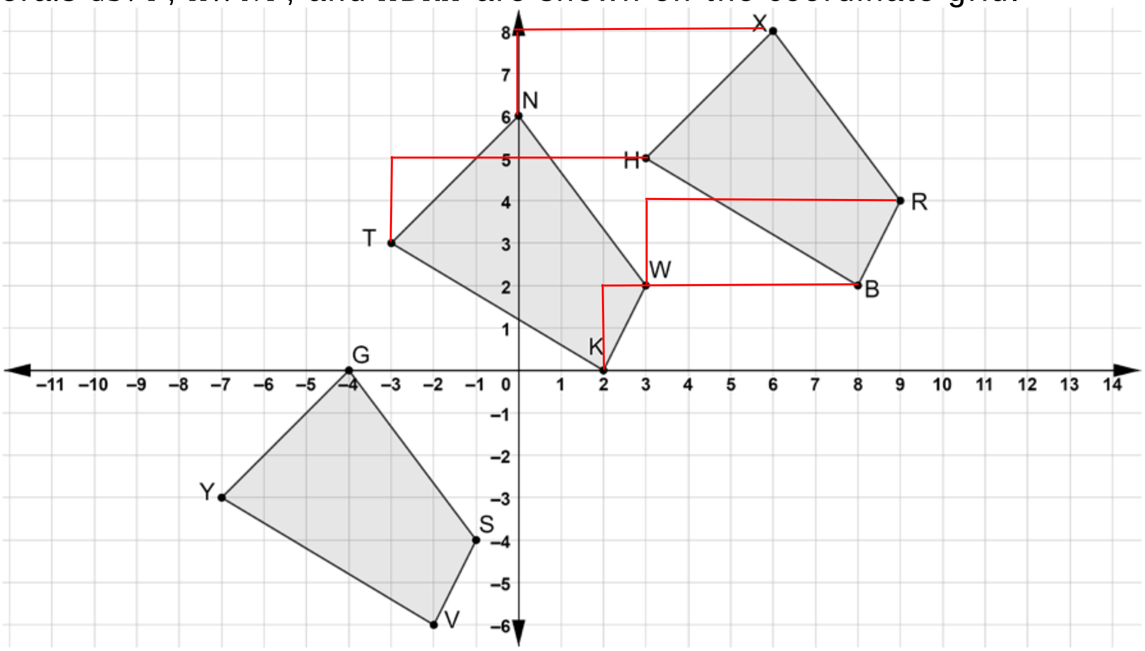
Geometry | Name_Answer Key_________
Unit 4 | Date_____________________
Lesson 1 Homework | Period___________________

Quadrilaterals $GSVY$, $KWNT$, and $HBRX$ are shown on the coordinate grid.



1. Draw the mapping that shows the translation if quadrilateral $TKWN$ is the preimage and quadrilateral $HBRX$ is the image.

2. Describe the translation if quadrilateral $TKWN$ is the preimage and quadrilateral $HBRX$ is the image.

   up 2, right 6

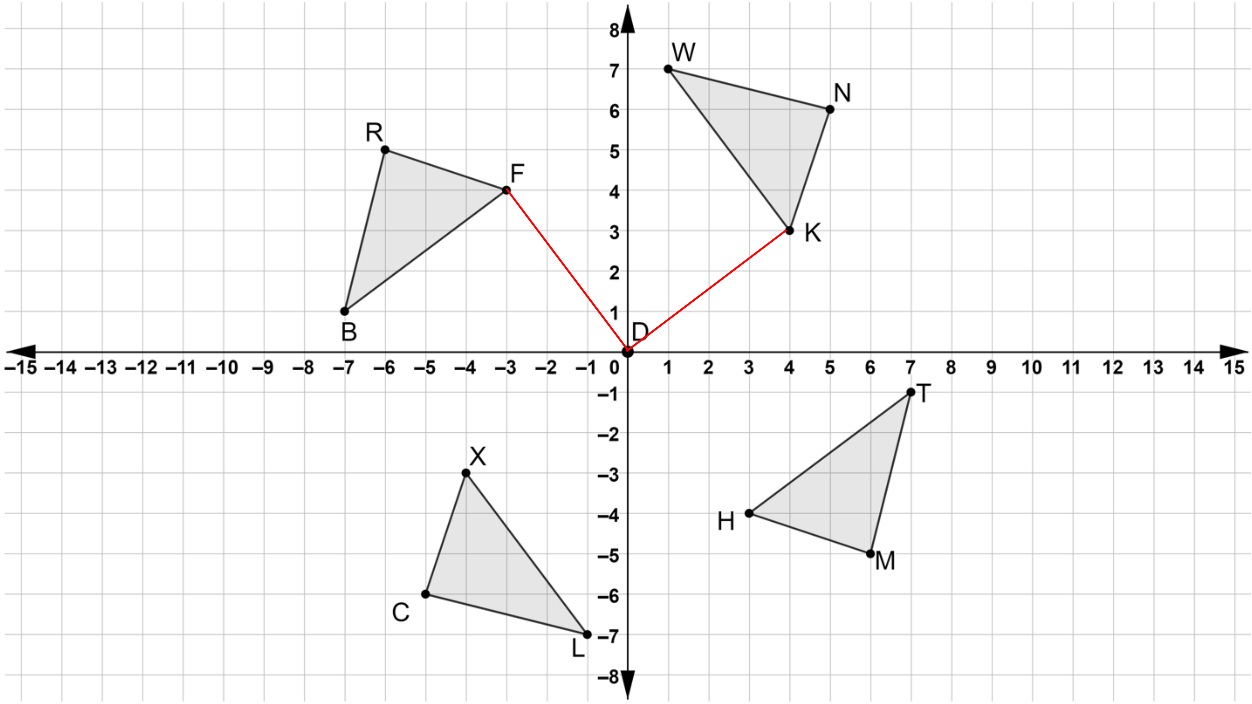
3. Complete the statements.
   The x-values of quadrilateral $GSVY$, the image, can be found by
   - o adding
   - ★ subtracting
   - o multiplying

   _10_ to the x-values of quadrilateral $XRBH$, the pre-image. The y-values of quadrilateral $GSVY$, the image, can be found by
   - o adding
   - ★ subtracting

   _8_ units to the y-values of quadrilateral $XRBH$, the pre-image.

4. Describe the translation if quadrilateral $XRBH$ is the preimage and quadrilateral $GSVY$ is the image.
   left 10, down 8

Four triangles are shown on the coordinate grid. Triangles WNK, TMH, and CXL are all rotations of $\triangle RFB$.



5. On the coordinate grid draw angle FDK.

6. Describe how the measure of $\angle FDK$ can be used to determine the degrees of rotation where $\triangle RFB$ is the pre-image and $\triangle NKW$ is the image.

   The angle measurement is the angle of rotation when the center of rotation is the origin.

7. Describe how the coordinates of the vertices of $\triangle RFB$, the preimage, and the coordinates of $\triangle NKW$, the image can be used to determine the degrees of rotation.
   How the coordinates change can indicate the angle of rotation.

8. Describe the rotation where $\triangle RFB$ is the preimage and $\triangle CXL$ is the image.

   One quadrant, 90° counterclockwise.

9. Describe how the coordinates of the vertices of $\triangle RFB$ and $\triangle TMH$ show that rotating $\triangle RFB$ $180°$ about the origin results in $\triangle MHT$.

$R(-6,5)$	$T(7,-1)$	
		Both x and y coordinates change sign
$F(-3,4)$	$M(6,-5)$	between preimage and image.
$B(-7,1)$	$H(3,-4)$	

10. Explain why the rotation of $\triangle RFB$ that results in $\triangle MHT$ can be described as $180°$ clockwise or $180°$ counterclockwise.

    Moving 180° counterclockwise is the same as moving 180° clockwise.

Geometry
Unit 4
Lesson 2 Homework

Name_Answer Key_________
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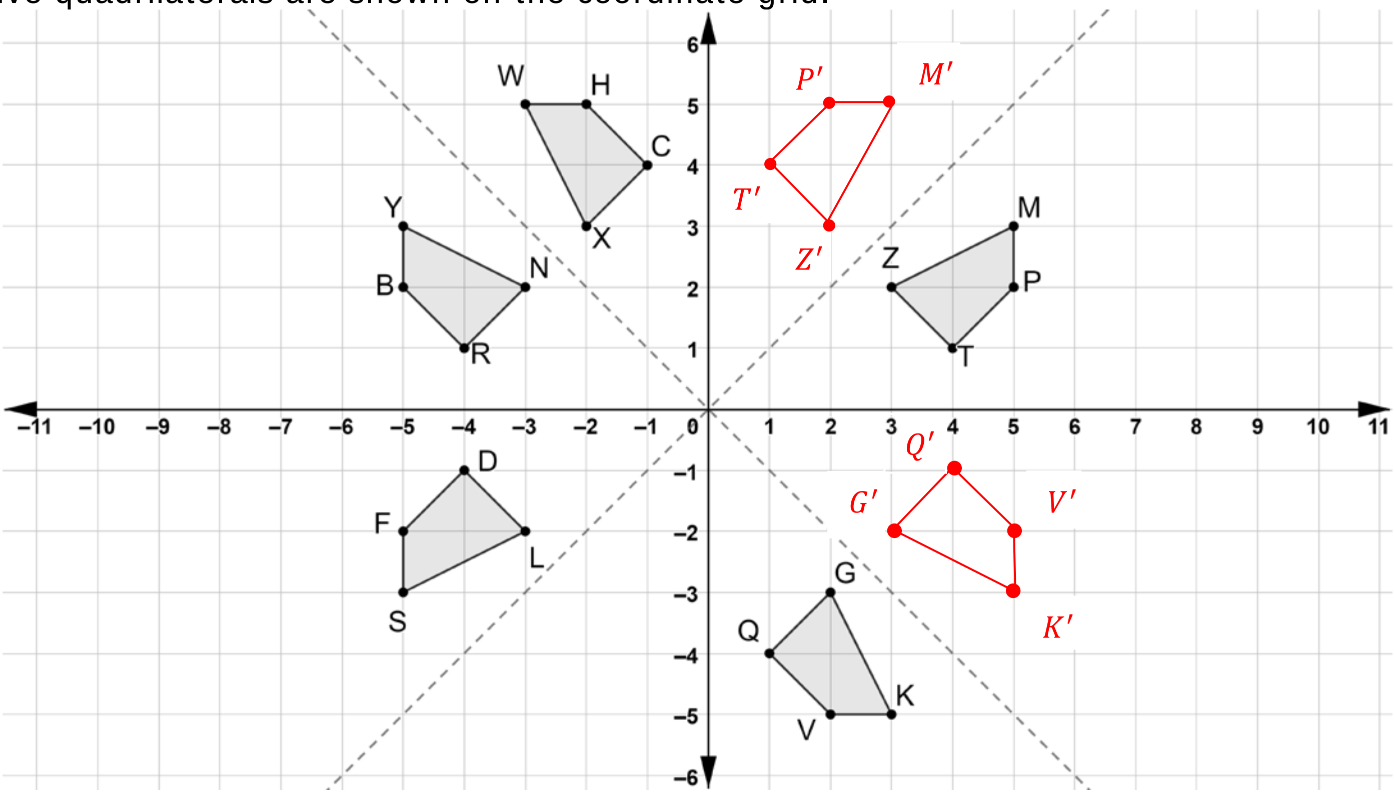
Five quadrilaterals are shown on the coordinate grid.



1. Draw the reflection of quadrilateral $MPTZ$ across the line $y = x$. Label the reflection as $M'P'T'Z'$.

2. Draw the reflection of quadrilateral $VKGQ$ across the line $y = -x$. Label the reflection as $V'K'G'Q'$.

3. Select all of the quadrilaterals that are reflections of quadrilateral $YBRN$.
   - ☐ Quadrilateral $WHCX$
   - ☐ Quadrilateral $MPTZ$
   - ☐ Quadrilateral $KVQG$
   - ☐ Quadrilateral $SFDL$
   - ☐ Quadrilateral $M'P'T'Z'$
   - ☐ Quadrilateral $K'V'Q'G'$

4. The algebraic description of the reflection of quadrilateral $BYNR$, the preimage, onto quadrilateral $PMZT$, the image, is $(x, y) \rightarrow (-x, y)$. Describe how the coordinates of the vertices of quadrilateral $BYNR$, the preimage, and the coordinates of quadrilateral $PMZT$, the image, can be used to determine the line of reflection.

$B: (-5, 2)$ $P: (5, 2)$
$Y: (-5, 3)$ $M: (5,3)$
$N: (-3, 2)$ $Z: (3,2)$
$R: (-4, 1)$ $T: (4, 1)$

The line of reflection is equidistant from the preimage coordinates to the image coordinates.

5. Which quadrilateral is the image of reflecting quadrilateral $DLSF$ across the x-axis?

$RNYB$

6. Which quadrilateral is the image of reflecting quadrilateral $M'P'T'Z'$ across the y-axis?

$WHCX$

7. Describe how the coordinates of the vertices of quadrilateral $BYNR$, the preimage, and the coordinates of quadrilateral $FSLD$, the image, can be used to determine the algebraic description of the reflection.

$B: (-5, 2)$	$F: (-5, -2)$	$(x, y) \to (x, -y)$
$Y: (-5, 3)$	$S: (-5, -3)$	
$N: (-3, 2)$	$L: (-3, -2)$	All x-coordinates are the same, the
$R: (-4, 1)$	$D: (-4, -1)$	y-coordinates are the opposite signs.

8. Describe how the coordinates of the vertices of quadrilateral $WHCX$, the preimage, and the coordinates of quadrilateral $YBRN$, the image, can be used to determine the algebraic description of the reflection.

$W: (-3, 5)$	$Y: (-5, 3)$	The coordinates change spots and signs.
$H: (-2, 5)$	$B: (-5, 2)$	$(x, y) \to (-y, -x)$
$C: (-1, 4)$	$R: (-4, 1)$	
$X: (-2, 3)$	$N: (-3, 2)$	

9. Write the algebraic description for the reflection of quadrilateral $BYNR$, the preimage, onto quadrilateral $PMZT$, the image.

$B: (-5, 2)$	$P: (5, 2)$	
$Y: (-5, 3)$	$M: (5, 3)$	x-coordinate changes signs.
$N: (-3, 2)$	$Z: (3, 2)$	$(x, y) \to (-x, y)$
$R: (-4, 1)$	$T: (4, 1)$	

10. Write the algebraic description for the reflection of quadrilateral $V'K'G'Q'$, the preimage, onto quadrilateral $HWXC$, the image.

V': $(5, -2)$	$H: (-2, 5)$	
K': $(5, -3)$	$W: (-3, 5)$	The x-coordinate and y-coordinate
G': $(3, -2)$	$X: (-2, 3)$	change spots.
Q': $(4, -1)$	$C: (-1, 4)$	$(x, y) \to (y, x)$

Geometry
Unit 4
Lesson 3 Homework

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Date___________________
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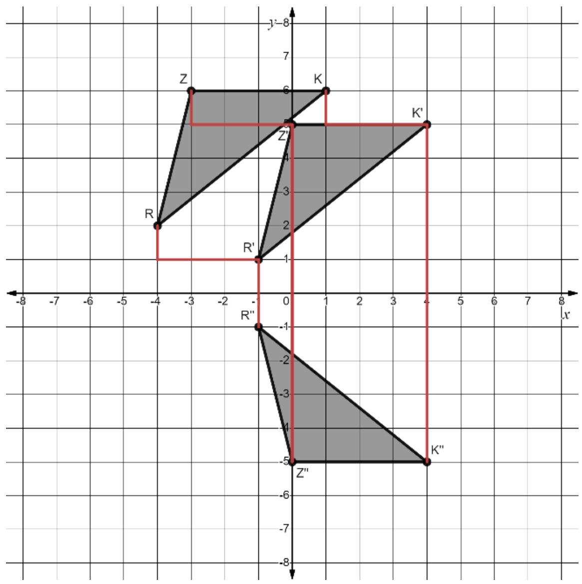
Triangles RZK, $R'Z'K'$, and $R''Z''K''$ are shown on the coordinate grid where $\triangle RZK$ is the preimage.



$\therefore R(-4,2) \rightarrow R'(-1,1)$

$Z(-3,6) \rightarrow Z'(0,5)$

$K(1,6) \rightarrow K'(4,5)$

$x+3, y-1$

$\therefore R'(-1,1) \rightarrow R''(-1,-1)$

$Z'(0,5) \rightarrow Z''(0,-5)$

$K'(4,5) \rightarrow K''(4,-5)$

$(x,y) \rightarrow (x,-y)$

1. Describe the feature of $\triangle R'Z'K'$ that indicates the rigid motion that was applied to $\triangle RZK$ that resulted in $\triangle R'Z'K'$. Answers will vary.
The location of the vertices are translated horizontally and vertically from the pre-image.
2. Show the feature described by mapping the transformation of one vertex of $\triangle RZK$ to $\triangle R'Z'K'$.

3. Complete the table.

Transformation	Mapping	Algebraic Description
Vertical translation down 1 unit. Horizontal translation right 3 units.	$\triangle RZK \rightarrow \triangle R'Z'K'$	$(x,y) \rightarrow (x+3, y-1)$

4. Map the transformation of one vertex of $\triangle R'Z'K'$ to $\triangle R''Z''K''$.

5. Complete the table.

Transformation	Mapping	Algebraic Description
Reflection across the x-axis.	$\triangle R'Z'K' \rightarrow \triangle R''Z''K''$	$(x,y) \rightarrow (x,-y)$

Triangles EFH and WSY are shown on the coordinate grid where $\triangle EFH$ is the preimage.

$\therefore E(-9,-5) \rightarrow W(3,3)$

$F(-6,2) \rightarrow S(0,-4)$

$H(-8,-8) \rightarrow Y(2,6)$

(x, y)

$\downarrow$

$(-x,-y) \rightarrow 180^\circ$

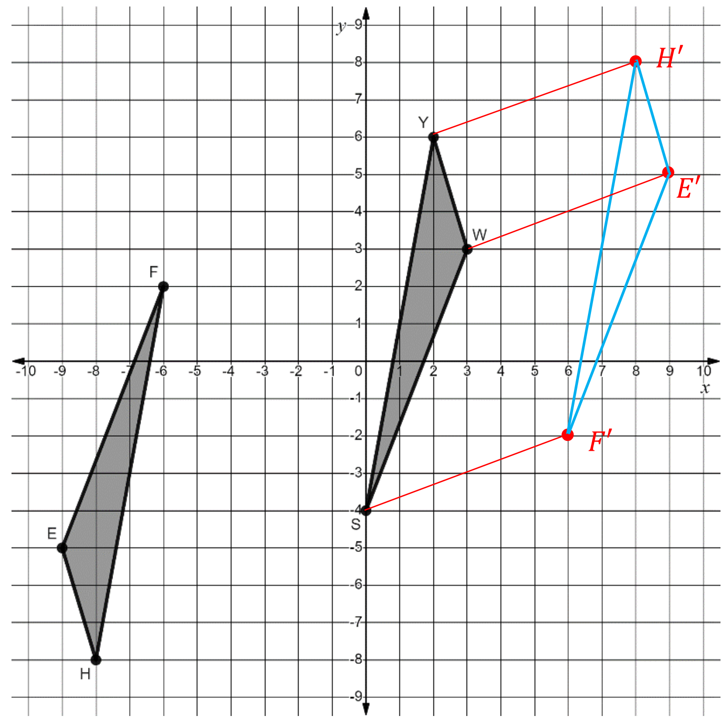
rotation $\overline{CW}$ about the origin

$(9,5)$

$(6,-2)$

$(8,8)$

$x-6, y-2$



6. Describe the feature of $\triangle WSY$ that indicates that more than one rigid motion was applied to $\triangle EFH$. Answers will vary.
The location of the vertices are rotated and then translated horizontally and vertically from the pre-image.

7. Draw $\triangle E'F'H'$, the result of the first transformation that was applied to $\triangle EFH$.

8. Map the transformation of one vertex of $\triangle E'F'H'$ to $\triangle WSY$.

9. Complete the table.

Transformation	Mapping	Algebraic Description
180° rotation about the origin	$\triangle EFH \rightarrow \triangle E'F'H'$	$(x, y) \rightarrow (-x,-y)$
Vertical translation down 2 units. Horizontal translation left 6 units	$\triangle E'F'H' \rightarrow \triangle WSY$	$(x, y) \rightarrow (x-6, y-2)$

10. Explain how you know if the sequence of transformations preserves distance.

Length of $\overline{EF} = \sqrt{58} \approx 7.6158$ Answers will vary.

Length of $\overline{E'F'} = \sqrt{58} \approx 7.6158$ Each of the vertices on the triangles are the same

Length of $\overline{WS} = \sqrt{58} \approx 7.6158$ distance apart after each transformation.

Geometry
Unit 4
Lesson 4 Homework

Name_Answer Key_________
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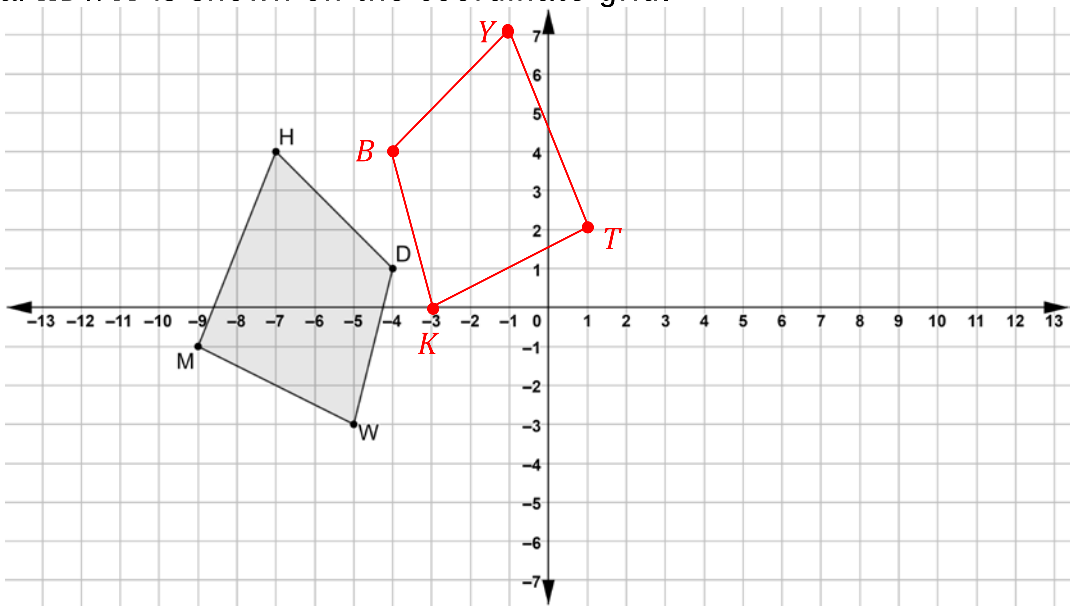
Quadrilateral $HDWM$ is shown on the coordinate grid.



1. Draw the quadrilateral that is the result of a translation right 8 units and up 3 units and then reflected across the y-axis. Label the new quadrilateral $YBKT$.

Complete the table.

	Transformation	Mapping	Algebraic Description
2.	translation	$HDWM \rightarrow H'D'W'M'$	$(x, y) \rightarrow (x + 8, y + 3)$
3.	reflection	$H'D'W'M' \rightarrow YBKT$	$(x, y) \rightarrow (-x, y)$

4. Complete the table of values for each quadrilateral.

Quadrilateral $HDWM$			Quadrilateral $H'D'W'M'$			Quadrilateral $YBKT$		
Vertex	x	y	Vertex	x	y	Vertex	x	y
H	-7	4	H'	1	7	Y	-1	7
D	-4	1	D'	4	4	B	-4	4
W	-5	-3	W'	3	0	K	-3	0
M	-9	-1	M'	-1	2	T	1	2

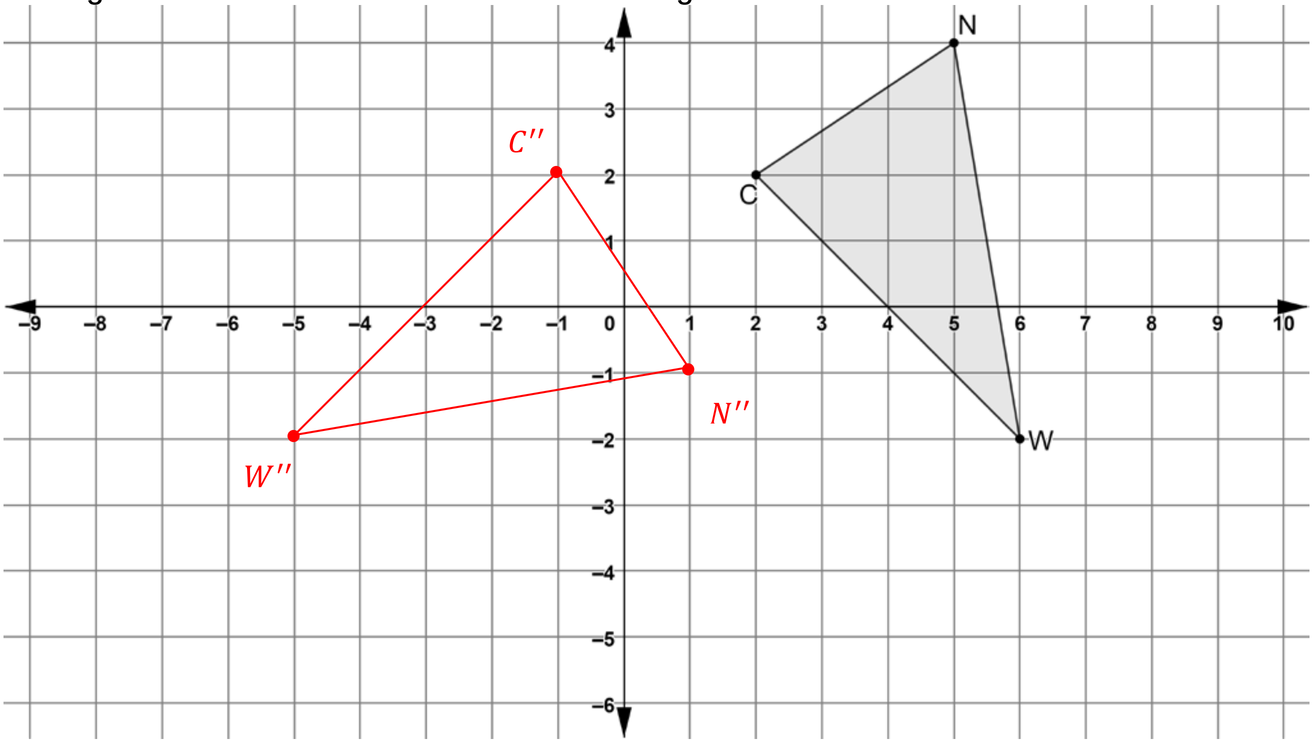
5. Explain how the tables for quadrilaterals $HDWM$ and $H'D'W'M'$ show the effect of the transformation on the coordinates that result in the algebraic description.

Each x-coordinate increases by 8, each y-coordinate increases by 3.

6. Explain how the tables for quadrilaterals $H'D'W'M'$ and $YBKT$ show the effect of the transformation on the coordinates that result in the algebraic description.

The x-coordinate changes its sign.

Triangle NWC is shown on the coordinate grid.



7. Draw the triangle that is the result of a rotation $90°$ clockwise about the origin and then a translation of left 3 units and up 4 units. Label the new triangle $N''W''C''$.

Complete the table.

	Transformation	Mapping	Algebraic Description
8.	rotation $90°$ clockwise	$\triangle NWC \to \triangle N'W'C'$	$(x, y) \to (y, -x)$
9.	translation	$\triangle N'W'C' \to \triangle N''W''C''$	$(x, y) \to (x - 3, y + 4)$

10. Describe the connection between the algebraic description of the rotation and the vertices of $\triangle NWC$ and $\triangle N'W'C'$.

The coordinates change spots and the second (now y-coordinate) changes sign.

Geometry
Unit 4
Lesson 5 Homework

Name Answer Key
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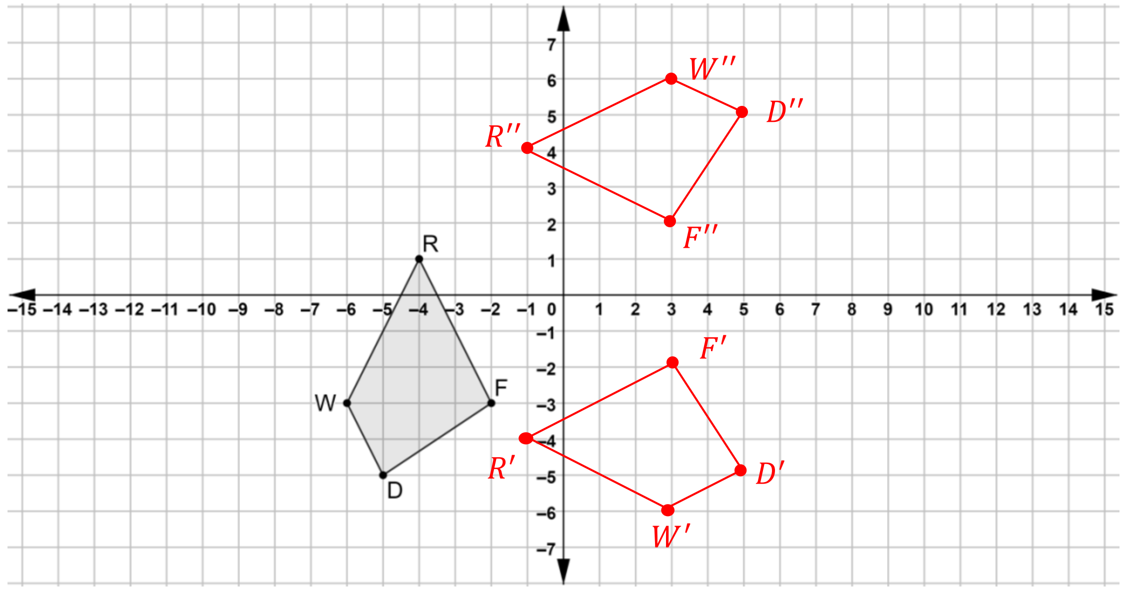
Quadrilateral $WRFD$ is shown on the coordinate grid.

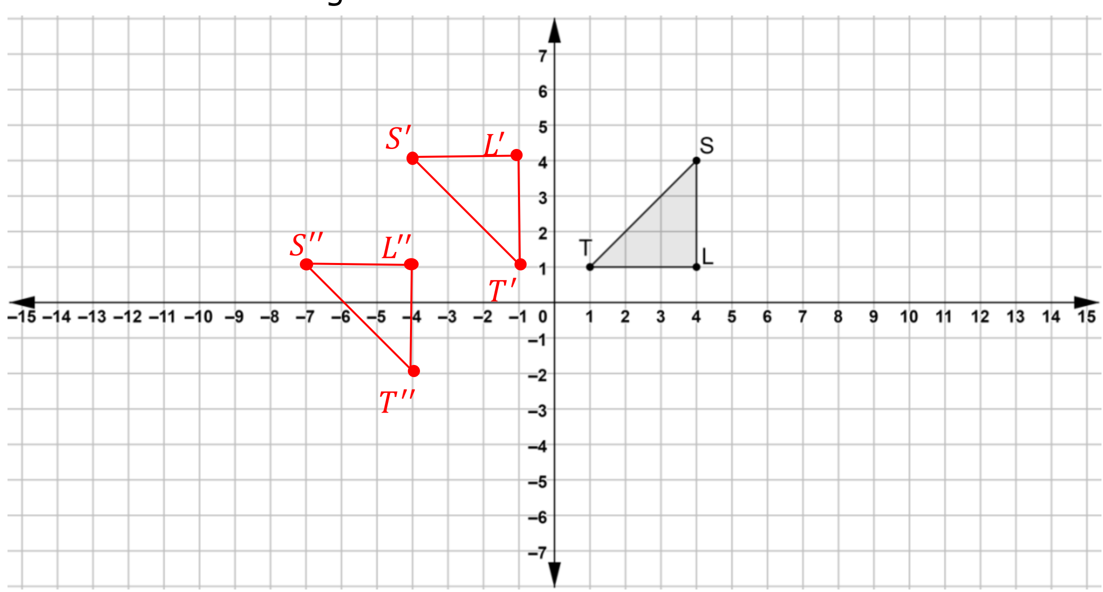
Perform the series of transformations on quadrilateral $WRFD$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Quadrilateral
1.	$(x, y) \rightarrow (-y, x)$	$W'R'F'D'$
2.	$(x, y) \rightarrow (x, -y)$	$W''R''F''D''$

Triangle TLS is shown on the coordinate grid.

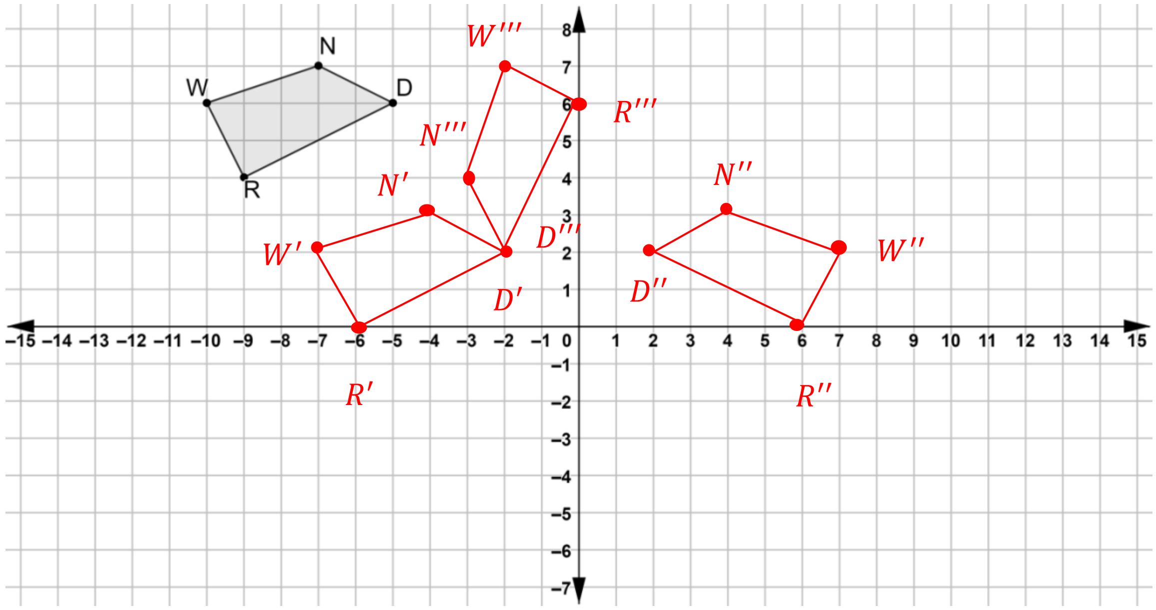
Perform the series of transformations on $\triangle TLS$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Triangle
3.	$(x, y) \rightarrow (-y, x)$	$T'L'S'$
4.	$(x, y) \rightarrow (x - 3, y - 3)$	$T''L''S''$

Quadrilateral $WNDR$ is shown on the coordinate grid.

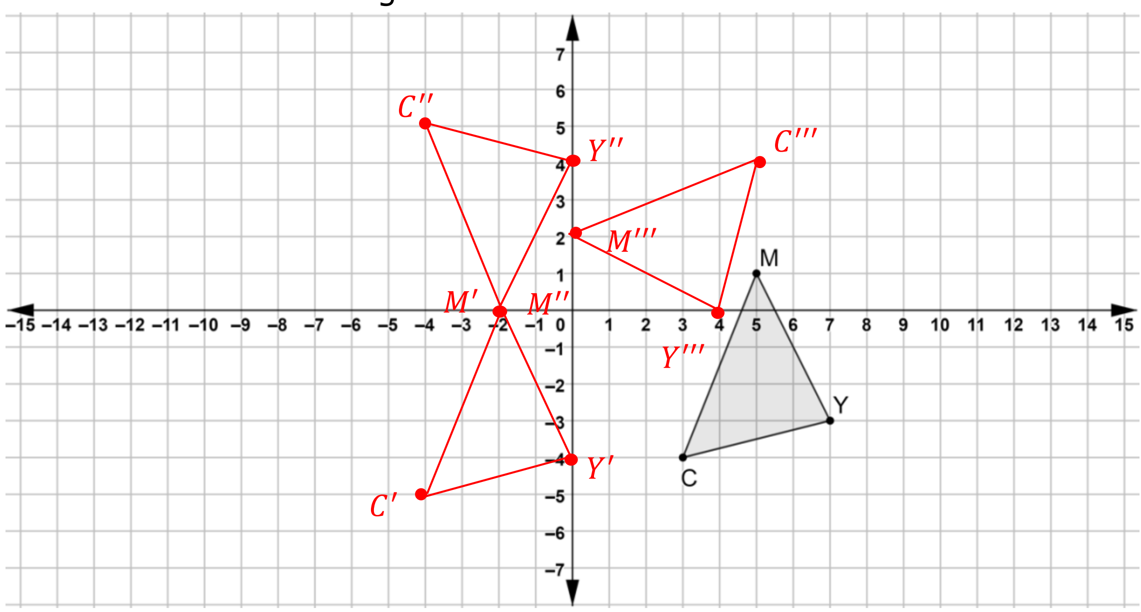
Perform the series of transformations on quadrilateral $WNDR$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Quadrilateral
5.	$(x, y) \rightarrow (x + 3, y - 4)$	$W'N'D'R'$
6.	$(x, y) \rightarrow (-x, y)$	$W''N''D''R''$
7.	$(x, y) \rightarrow (-y, x)$	$W'''N'''D'''R'''$

Triangle MYC is shown on the coordinate grid.

Perform the series of transformations on $\triangle MYC$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Triangle
8.	$(x, y) \rightarrow (x - 7, y - 1)$	$M'Y'C'$
9.	$(x, y) \rightarrow (x, -y)$	$M''Y''C''$
10.	$(x, y) \rightarrow (y, -x)$	$M'''Y'''C'''$

Geometry
Unit 4
Lesson 6 Homework

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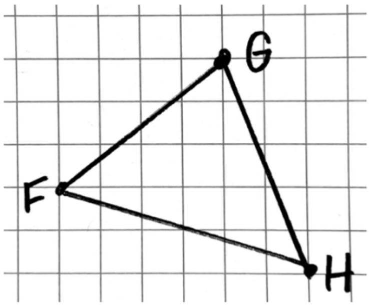
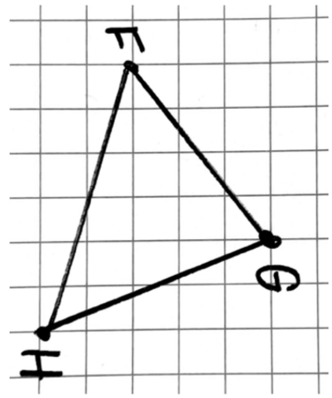
Tamika drew $\triangle FGH$ on graph paper. She then rotated the graph paper and moved it to the right.





1. Explain what method(s) can be used to determine if each side of $\triangle FGH$ has the same length as the triangle that has been rotated and moved.
   Patty paper can be used to copy each side of $\triangle FGH$ or a ruler can be used to measure each side of $\triangle FGH$.

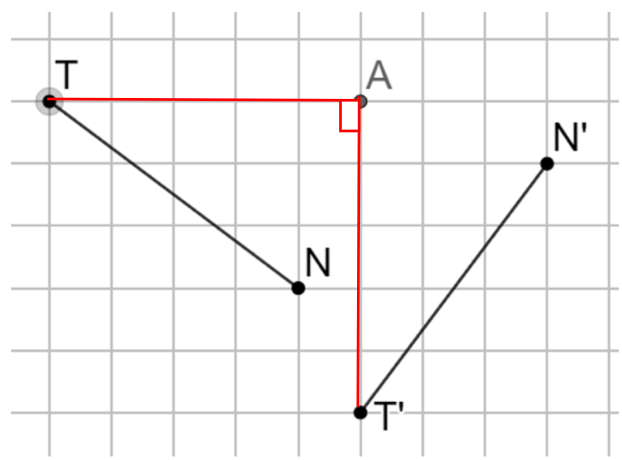
2. Explain what method(s) can be used to determine if each angle of $\triangle FGH$ has the same measure as the triangle that has been rotated and moved.
   Patty paper can be used to copy each angle of $\triangle FGH$ or a protractor can be used to measure each angle of $\triangle FGH$.

3. Explain why showing that the side lengths and the angles have the same measure shows that the transformations used can be classified as isometry.

   Because angle measures and side lengths are preserved.

Line segments TN, $T'N'$, and point A are shown on the grid.

4. Draw $\angle TAT'$. Label the angle measure.
   90°



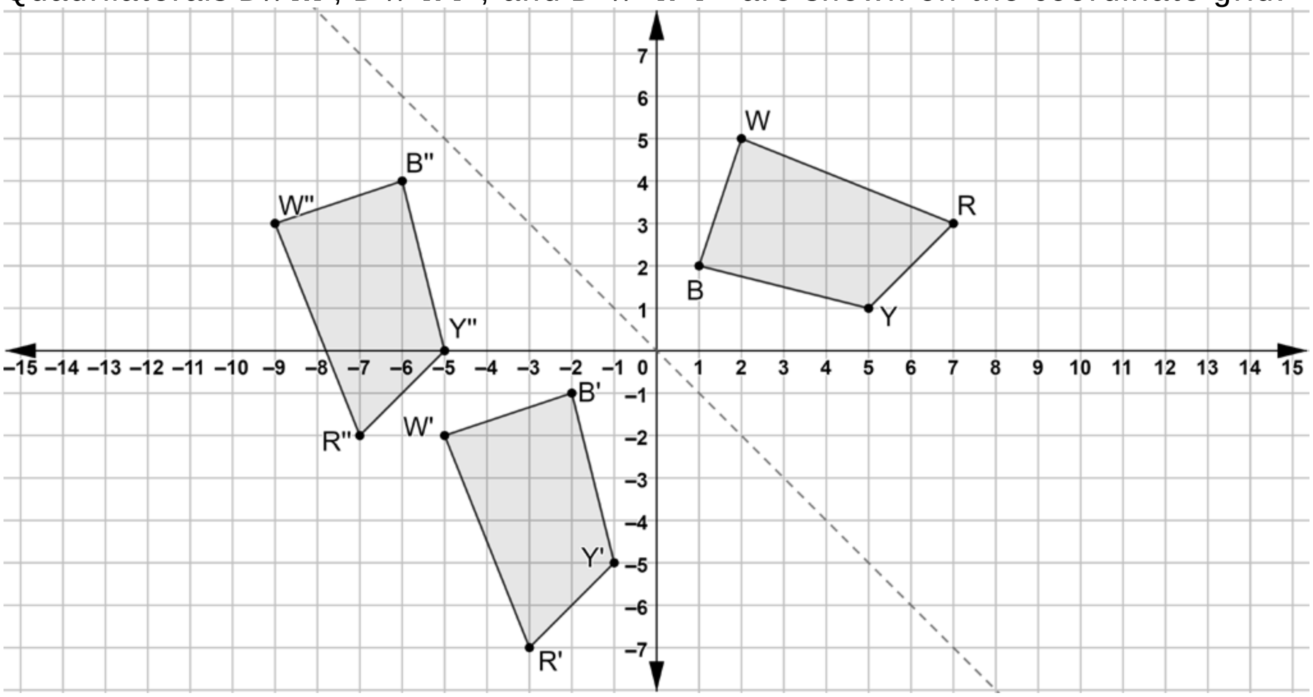
5. Describe the transformation applied to $\overline{TN}$ that resulted in $\overline{T'N'}$.

   Rotate 90° counterclockwise about Point A.

6. Complete the statement.

   There is a(n) ______isometry_______ between the set of points on $\overline{TN}$ and the set of points on $\overline{T'N'}$, therefore all of the points on ____$\overline{TN}$____ have to have a matching point on ____$\overline{T'N'}$____.

Quadrilaterals $BWRY$, $B'W'R'Y'$, and $B''W''R''Y''$ are shown on the coordinate grid.



7. Describe what movements should be done with a patty paper copy of quadrilateral $BWRY$ so that the copy of $BWRY$ lays on top of $B'W'R'Y'$.

   Reflection over the line $y = -x$.

8. Describe what movements should be done with a patty paper copy of quadrilateral $B'W'R'Y'$ so that the copy of $B'W'R'Y'$ lays on top of $B''W''R''Y''$.

   Translation left 4 units, up 5 units.

9. Complete the statements.
   The transformations used in both sequences

   - o are
   - o are not

   examples of isometry. All of the points of

   - o $B'W'R'Y'$
   - o $B''W''R''Y''$
   - o $B'W'R'Y'$ and $B''W''R''Y''$

   can be mapped to all of the points of $BWRY$.

10. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $BWRY$ resulted in $BWRY \cong B''W''R''Y''$.

    In both sequences of transformations only rigid motions were used to map $BWRY$ to $B'W'R'Y'$ to $B''W''R''Y''$ so the figures are congruent by the definition of congruence in terms of rigid motions.

Geometry
Unit 4
Lesson 7 Homework

Name_Answer Key_________
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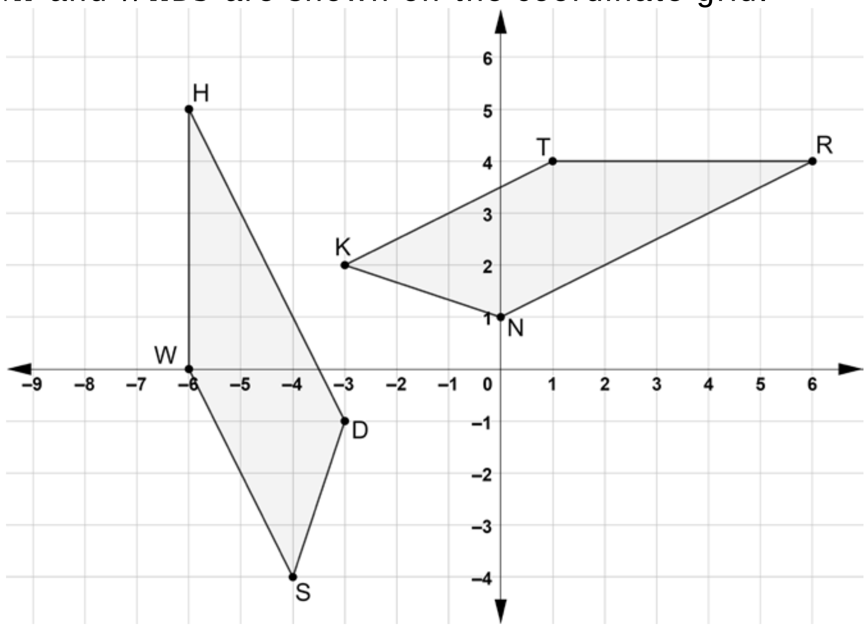
Quadrilaterals $TRNK$ and $WHDS$ are shown on the coordinate grid.



1. Describe a sequence of transformations that will map quadrilateral $TRNK$ onto quadrilateral $WHDS$.
   Rotate 90° counterclockwise about the origin, translate left 2 units and down 1 unit.

2. The sequence of transformations that will map quadrilateral $TRNK$ onto quadrilateral $WHDS$ includes a rotation. Describe the sequence using a different rotation.
   Rotate 270° clockwise about the origin, translate left 2 units and down 1 unit.

3. Complete the table that shows which vertices were mapped to each other when quadrilateral $TRNK$ was mapped to quadrilateral $WHDS$.

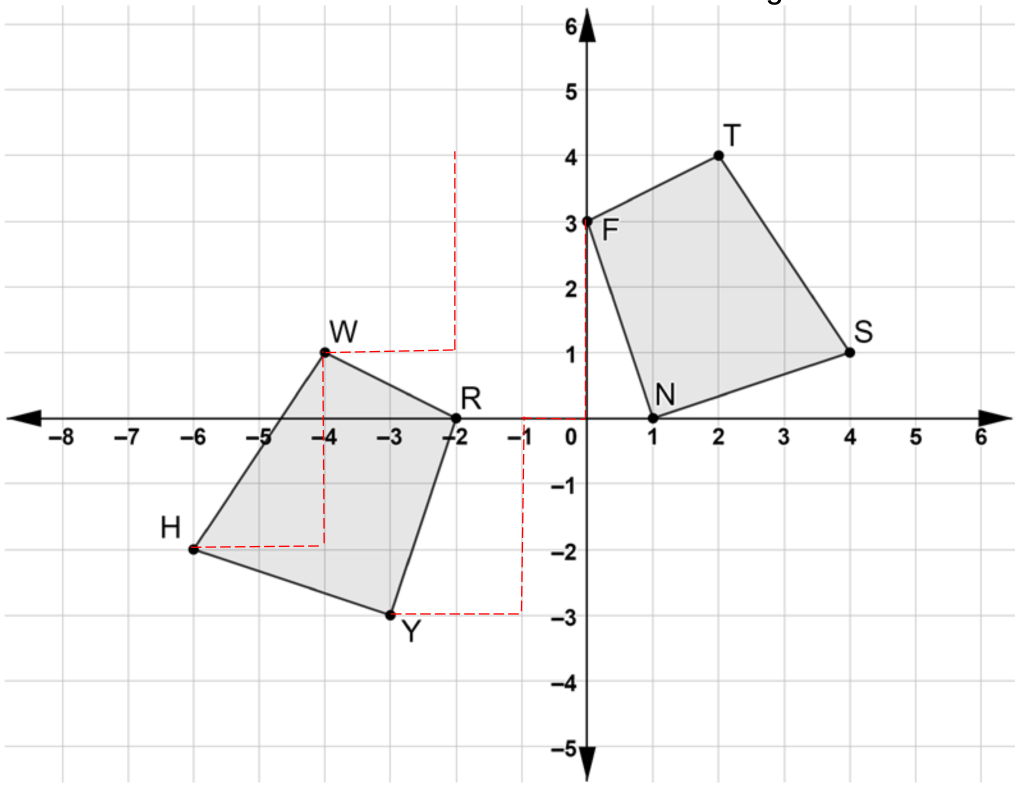
Quadrilateral $TRNK$	**Quadrilateral $WHDS$**
T	W
R	H
N	D
K	S

4. Write the algebraic description that matches the written description one of the transformations.

$$(x, y) \rightarrow (-y, x)$$
$$(x, y) \rightarrow (x - 2, y - 1)$$

5. Complete the congruence statement. $TRNK \cong WHDS$

Quadrilaterals $YHWR$ and $FTSN$ are shown on the coordinate grid.



6. Describe a sequence of transformations that will map quadrilateral $YHWR$ onto quadrilateral $FTSN$.

   Translate right 2 units and up 3 units, then reflect across the y –axis.

7. Show the mapping of the transformations on the coordinate grid.

8. Complete the statement.

   Vertex Y maps to vertex N, vertex W maps to vertex T, vertex H maps to vertex S, and vertex R maps to vertex F.

9. Write the algebraic description that matches the written description of the transformations.

$$(x, y) \rightarrow (x + 2, y + 3)$$
$$(x, y) \rightarrow (-x, y)$$

10. Complete the congruence statement. $WRYH \cong TFNS$

Geometry
Unit 4
Lesson 8 Homework

Name_Answer Key_________
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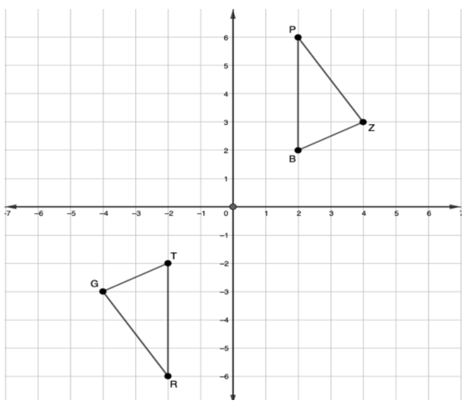
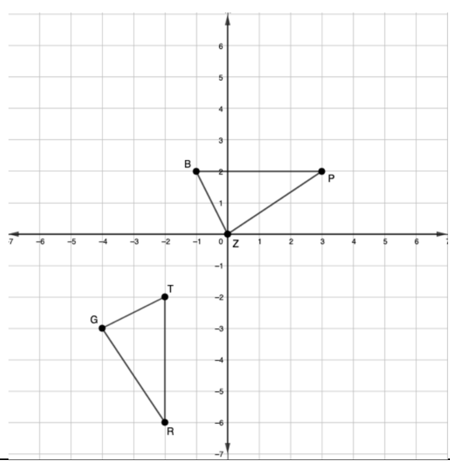
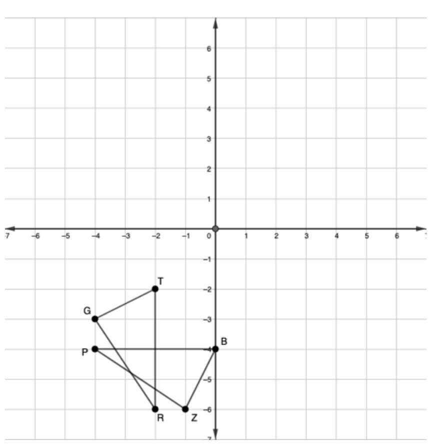
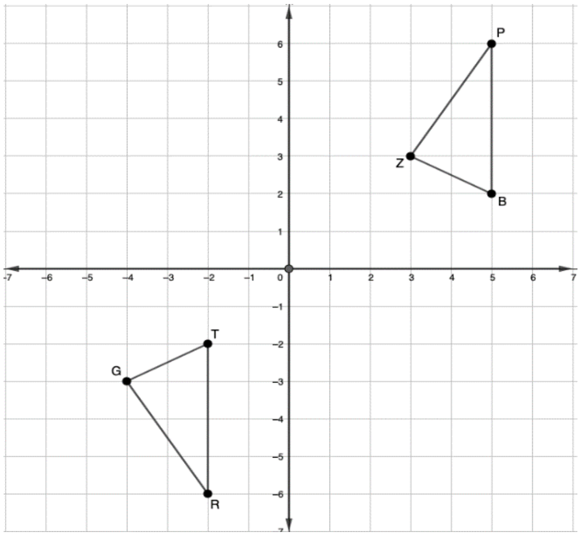
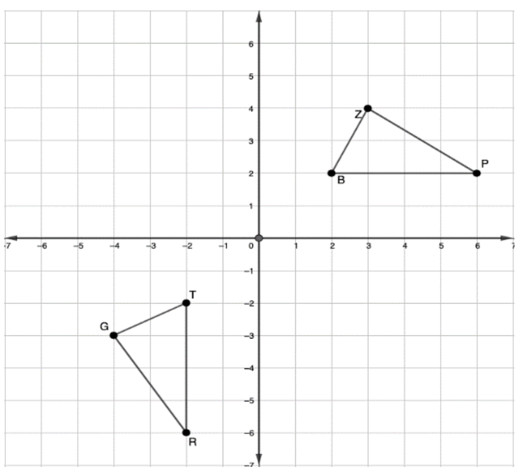
Period___________________

For each coordinate grid with the pre-image of $\triangle TGR$ and image $\triangle BZP$, where $\triangle TGR \cong \triangle BZP$. Complete the table with the missing written and algebraic description(s) that map $\triangle TGR$ onto $\triangle BZP$.

Coordinate Grid	Written Description	Algebraic Description
	1. 180° rotation about the origin.	2. $(x, y) \rightarrow (-x, -y)$
	3. Translate right 4 units and up 3 units, then rotate 90° counterclockwise about the origin.	4. $(x, y) \rightarrow (x+4, y+3)$ $(x, y) \rightarrow (-y, x)$
	5. Reflect over the line $y = x$, then translate right 2 units and down 2 units.	6. $(x, y) \rightarrow (y, x)$ $(x, y) \rightarrow (x+2, y-2)$

	7. Translate right 7 units, then reflect over the x-axis.	8. $(x, y) \rightarrow (x + 7, y)$ $(x, y) \rightarrow (x, -y)$
	9. Rotate 90° clockwise, then reflect over the y-axis.	10. $(x, y) \rightarrow (y, -x)$ $(x, y) \rightarrow (-x, y)$